

Gradient-Aware Adaptive Rounding for Post-Training Quantization of Deep Neural Networks

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The recent exponential scaling of Large Language Models (LLMs) has resulted in massive carbon emissions and significant challenges in deployment, forcing users to rely on third-party cloud providers, thereby raising privacy concerns. This highlights the necessity of model compression, and post-training quantization (PTQ) is one method that enables efficient on-device models. AdaRound, an adaptive rounding method that outperforms round-to-nearest rounding during quantization, minimizes the task loss between the original weights and the quantized weights on unlabelled calibration data. We find that: 1) AdaRound leaks some information about its calibration data into the weight perturbation matrix, and we demonstrate this by proposing a novel inversion attack that targets the null space of this matrix to recover coarse representations of the original data. 2) The theoretical foundation of AdaRound asserts that the gradient of the loss function is zero for pretrained converged models, an assumption which is refuted by recent literature. To address this discrepancy, we attempted to reconcile this by incorporating local gradient signals into the AdaRound optimization process, but obtained neutral results on empirical evaluations. Therefore, this thesis highlights a frequently overlooked dimension in PTQ (privacy), and proposes a gradient-aware modification of AdaRound to better align the method with modern literature.

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Exponential Scaling of LLMs

Recently, the exponential scaling of LLM sizes has inadvertently resulted in models that are extremely challenging to deploy, thus causing users to rely on third-party providers for LLM usage.

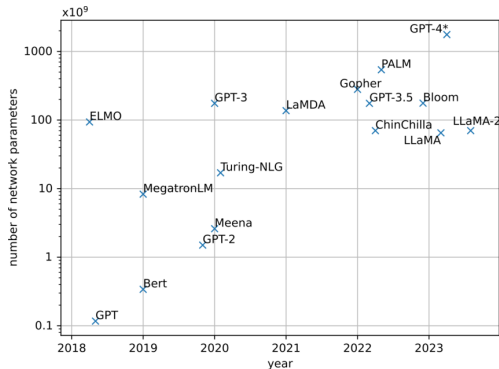


Figure: Observed exponential scaling of LLM sizes over time [1] (Note: The GPT-4 figure is an estimate)

Why Compress LLMs?

- ▶ Environmental Effects: GPT-3's training had an estimated CO₂e of ~ 552.1 tons [2], GPT-4's training had an estimated CO₂e of ~ 6912 tons [3], and Llama 3.3 [4]'s training had an estimated CO₂e of $\sim 11,390$ tons.
- ▶ Cost: For example, OpenAI o1 [5] in “pro mode” costs \$150 per 1 million tokens input, and \$600 per 1 million tokens output. Uber's yearly AI budget was exhausted in 4 months, while Microsoft began cancelling Claude Code licenses just 6 months after acquiring them [6].
- ▶ Reliability. Single point of failure, and an outage can have severe effects on mission-critical tasks. For example, it is observed that whenever ChatGPT has an outage, stock trading volume goes down [7].

Benefits of On-Device Inference

- ▶ Privacy: Data does not need to leave the device
- ▶ Robustness: Can function even without internet access
- ▶ Latency: Doesn't have to rely on third-party scheduling, can prioritize urgent tasks
- ▶ Cost: Reduces API fees, network costs, and even environmental emissions

Model Compression Methods

Methods typically used for the compression of deep neural networks:

- ▶ Weight Pruning: Setting some weights to zeroes, resulting in sparse weight matrices [8].
- ▶ Knowledge Distillation: Smaller model learns to match output logits of larger model [9].
- ▶ Weight Palettization: Clusters weights (e.g. via k -means), then stores every weight as the centroid of the cluster they belong to [10].
- ▶ Quantization: **We focus on this method for our thesis.** Stores weights in a data type of less bit width (e.g. int8), compared to the original data type (e.g. float32) [11].

Contributions

Our contributions in this work are twofold:

- ▶ First, we demonstrate a novel inversion method against AdaRound to recover coarse outlines that represent its calibration data, motivating the need to consider privacy in data-dependent quantization method.
- ▶ Second, we propose a gradient-aware adaptive rounding method based on AdaRound, and apply the proposed method on two language models.

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Compressibility of Deep Neural Networks

Source coding theorem [12]:

- ▶ n i.i.d. (independent and identically distributed) random variables, each with entropy $H(X)$, can be compressed to $nH(X)$ bits on average, with negligible information loss, as $n \rightarrow \infty$

In this case, $H(X)$, the entropy of a random variable, is the expected amount of information needed to describe it, and can be computed by:

$$H(X) := - \sum_{x \in X} p(x) \log_2 p(x). \quad (1)$$

Note that the log base determines the units: $\log_2$ yields bits, $\log_{10}$ yields dits, and $\ln$ yields nats.

Measuring Compressibility of LLMs

Total information stored in LLM weights: $\sim n \cdot \hat{H}(W)$ bits;
Maximum possible total entropy of the model: $n \cdot b$ bits. It therefore follows that the capacity utilization by a model is:

$$U(W) := \frac{\hat{H}(W)}{b}. \quad (2)$$

We use this method to quantify the capacity utilization of several modern LLMs:

Table: Entropy Estimate of Language Models (bfloat16)

Model	$\hat{H}(W)$ in bits	$U(W)$
Qwen 3 4B [13]	10.58	$10.58/16 \approx 66\%$
Qwen 3.5 4B [14]	10.56	$10.56/16 \approx 66\%$
Gemma 4 12B Instruct [15]	10.69	$10.69/16 \approx 67\%$
GLM 4.7 Flash [16] (30B)	10.49	$10.49/16 \approx 66\%$

Small Language Models

TinyStories [17] is a work that proposed a synthetic low-vocabulary dataset and several small language models, as low as 1 million parameters, that produce coherent English.

One limitation we observed in this work is their usage of the default GPT-Neo [18] tokenizer (which is identical to the pretrained GPT-2 [19] tokenizer), restricted to the top 10,000 tokens.

As a preliminary experiment, we train two small language models on TinyStories v1 and TinyStories v2, respectively. We use a SentencePiece Byte-Pair Encoding (BPE) tokenizer [20] with a vocabulary size of 1024, and we used a LLaMa-based [21] model with roughly half the parameters of the smallest reference model.

Evaluation of Our LMs

We use perplexity [22] as a quantitative metric to evaluate a language model's accuracy. For a model θ , a text x of length N , it is defined as the following.

$$\text{Perplexity}(\theta, x) = \left(\prod_{i=1}^N \frac{1}{p_{\theta}(x_i \mid x_1, x_2, \dots, x_{i-1})} \right)^{\frac{1}{N}} \quad (3)$$

In other words, it is the geometric mean of the inverse probabilities of a given text.

Table: Evaluation of Our Language Models

	Reference	Ours (TinyStories v1)	Ours (TinyStories v2)
Architecture	GPT-Neo	LLaMa	LLaMa
Parameter Count	1.17M	0.57M	0.46M
Training Data	TinyStories v1		TinyStories v2
Tokenizer (Vocab Size)	GPT-Neo	SentencePiece	SentencePiece
Vocab Size	10000	1024	1024
Perplexity ↓ (TinyStories v1)	19.3919	11.9006	—
Perplexity ↓ (TinyStories v2)	—	—	8.0994

Baseline Round-to-Nearest Quantization

Consider the symmetric uniform quantization method to int8:

- ▶ Quantization:

- ▶ $\text{scale} = \frac{\max(|W|)}{2^7 - 1}$

- ▶ $W_{\text{quantized}} = \text{clamp}(\lfloor \frac{W}{\text{scale}} \rceil)$

- ▶ Reconstruction (dequantization):

- ▶ $W_{\text{reconstructed}} = \text{scale} \cdot W_{\text{quantized}}$

- ▶ Note that $\text{clamp}()$ here refers to clamping a number between $[-2^7, 2^7 - 1]$ to ensure its within the range of the data type.

We can see that the model weights are perturbed in the direction of $W_{\text{reconstructed}} - W$ after quantization and dequantization. Let us use our TinyStories v2 model to investigate the effects of perturbation direction on the model's accuracy.

Loss Landscape: Direction Matters

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AdaRound: Objective

AdaRound [23] is a novel rounding method, which aims to minimize the difference in task loss of the quantized model and the unquantized model. The task loss of layer j of the unquantized model is $\mathcal{L}(x, y, w_j)$, and the task loss of the same layer of the quantized model $\mathcal{L}(x, y, w_j + \Delta w_j)$. Therefore, the objective is:

$$\arg \min_{\Delta w_j} (\mathcal{L}(x, y, w_j + \Delta w_j) - \mathcal{L}(x, y, w_j)). \quad (4)$$

Using second order Taylor expansion:

$$\mathcal{L}(x, y, w_j + \Delta w_j) - \mathcal{L}(x, y, w_j) \approx \nabla \mathcal{L}(x, y, w_j)^T \Delta w_j + \frac{1}{2} \Delta w_j^T \mathbf{H} \Delta w_j \quad (5)$$

AdaRound: Approximations

The authors assert that $\nabla \mathcal{L}(x, y, w_j)$ is negligible, which transforms the objective into just:

$$\mathcal{L}(x, y, w_j + \Delta w_j) - \mathcal{L}(x, y, w_j) \approx \frac{1}{2} \Delta w_j^T \mathbf{H} \Delta w_j \quad (6)$$

Finally, the authors formulate the optimization of weigh rounding for layer j as the following Quadratic Unconstrained Binary Optimization (QUBO) problem.

$$\arg \min_{\Delta w_j} \Delta w_j^T \mathbf{H} \Delta w_j \quad (7)$$

The above problem is NP-hard, so AdaRound uses the local mean squared error alongside a regularizer as a proxy.

$$\arg \min_{\mathbf{V}} \left(\|\mathbf{W}_X - \hat{\mathbf{W}}_X\|_F^2 + f_{\text{reg}}(\mathbf{V}) \right) \quad (8)$$

AdaRound: Algorithm

Algorithm 1 AdaRound

Require: $\mathbf{W}$, X (calibration data), t (number of iterations), b (target bit width), β (regularization parameter)

Ensure: Quantized model $\mathbf{W}_Q$, scales

- 1: $x_{in} \leftarrow \text{CACHEACTIVATIONS}(\mathbf{W}, X)$ ▷ Forward pass on calibration data to cache input activations
 - 2: **for** w_j in $\mathbf{W}$ **do**
 - 3: $q_{\min} \leftarrow -2^{b-1}$
 - 4: $q_{\max} \leftarrow 2^{b-1} - 1$
 - 5: $\text{scales}_j \leftarrow \frac{\max(w_j)}{q_{\max}}$ ▷ Store scale for output
 - 6: $\mathbf{V}_j \leftarrow h^{-1}(w_j - \lfloor w_j \rfloor)$ ▷ Initialize $\mathbf{V}_j$
 - 7: **for** $i \in \{0 \dots t\}$ **do**
 - 8: $\beta \leftarrow \text{ANNEAL}(\beta)$
 - 9: $\hat{w}_j \leftarrow \text{clamp} \left(\lfloor \frac{w_j}{\text{scales}_j} \rfloor + h(\mathbf{V}_j), q_{\min}, q_{\max} \right)$
 - 10: $\mathcal{L} \leftarrow \|w_j x_{in} - \hat{w}_j x_{in}\|_F^2 + \sum_{i,j} (1 - |2h(\mathbf{V}_{i,j}) - 1|^\beta)$
 - 11: $\mathbf{V} \leftarrow \mathbf{V} - \nabla \mathcal{L}$ ▷ via autograd
 - 12: **end for**
 - 13: $\mathbf{W}_{Q_{i,j}} \leftarrow \begin{cases} \lceil w_{i,j} \rceil, & \text{where } h(\mathbf{V}_{i,j}) \geq 0.5 \\ \lfloor w_{i,j} \rfloor, & \text{otherwise} \end{cases}$ ▷ Finalize quantization
 - 14: **end for**
 - 15: **return** $\mathbf{W}_Q$, scales
-

Limitations of AdaRound

- ▶ First, using calibration data can cause some information to be encoded into the weight perturbation matrix. We empirically verify this in Contribution 1.
- ▶ Second, the assumption in Equation 5 that the gradient of the loss function for pretrained converged models is negligible (for which empirical evidence was not provided) is challenged by recent literature.

AdaRound's Zero-Gradient Assumption

While one might expect gradient descent to follow a smooth, stable path to an optimum, in reality it almost always operate in a highly dynamic, oscillatory regime described as the **Edge of Stability** [24], leading to chaotic loss curves such as:

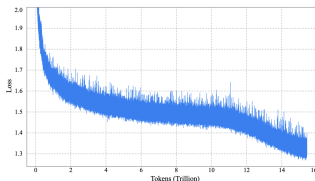


Figure: Training Loss Curve of Kimi K2 [25]

The oscillatory optimization also implies that the gradient for pretrained models are rarely zero. Recent literature [24] empirically show that even as the training loss decreases, the gradient norm of a model almost stays constant.

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Leakage Channel: The Null Space

- ▶ If we look at the AdaRound objective, the goal is to make $\|Wx - \hat{W}x\|_F^2$ approach zero. If we define the weight perturbation as $\Delta W = W - \hat{W}$, this means that for a model quantized using AdaRound:

$$\Delta Wx \approx 0$$

- ▶ Mathematically, this implies that the calibration data x is approximately in the **null space** of the perturbation matrix. In other words, $x \in \mathcal{N}(\Delta W)$. We can therefore exploit this property to generate samples that resemble the mean of the calibration data.
- ▶ We **can't** reconstruct exact pixel-for-pixel calibration samples. However, we demonstrate that by implementing this as a simple optimization problem, we can reconstruct representations that look like the mean of these calibration samples.

Realizing an Attack

- ▶ If an attacker has access to both the original weights (W) and the quantized weights ($\hat{W}$), they can recover the calibration data by searching for any vector that satisfies this null space property.
- ▶ For example, hypothetically: a company downloads an open-source vision model, quantizes it with AdaRound using confidential internal data (e.g. patient records), then deploys it onto edge devices sold to customers. An attacker can reconstruct representations of the calibration data by extracting the quantized $\hat{W}$ from the edge devices, then performing the inversion on the weight perturbation matrix $\Delta W = W - \hat{W}$.

Proposed Inversion Algorithm

Algorithm 2 AdaRound Inversion Algorithm

Require: W , $\hat{W}$, $\hat{x}_{\text{init}}$, No. of Iterations K , Learning Rate η

1: $\Delta W \leftarrow W - \hat{W}$

2: $\hat{x} \leftarrow \hat{x}_{\text{init}}$

3: **for** $k = 1$ **to** K **do**

4: $\mathcal{L} \leftarrow \sum (\Delta W \tanh(\hat{x}))^2$ (Forward pass)

5: $\frac{\partial \mathcal{L}}{\partial \hat{x}} \leftarrow \text{Autograd}(\mathcal{L}, \hat{x})$ (Backward pass)

6: $\hat{x} \leftarrow \hat{x} - \eta \frac{\partial \mathcal{L}}{\partial \hat{x}}$ (Update $\hat{x}$)

7: **end for**

8: **return** $\tanh(\hat{x})$

Experiment Setup

- ▶ We attempt an inversion on MLPs trained on the Olivetti faces dataset [26] the Weizmann Horse Database[27], and a Chest X-Ray Images dataset[28] using an MLP classifier quantized by AdaRound.
- ▶ For each model, we performed 500 iterations of AdaRound quantization. We then attempted to invert the process by training 10 random Gaussian samples for 3,000 iterations using the Adam optimizer. We report SSIM, PSNR, and Normalized RMSE before and after the inversion.

Results

Table: Metrics from Random Initialization vs. Our Method

	SSIM $\uparrow$	PSNR $\uparrow$	NRMSE $\downarrow$
Olivetti Faces (Before)	0.00	3.50	1.16
Olivetti Faces (After)	0.14 (+0.14)	12.83 (+9.33)	0.40 (-0.76)
Chest X-Ray (Before)	0.00	2.61	1.10
Chest X-Ray (After)	0.07 (+0.07)	10.13 (+7.52)	0.46 (-0.64)
Horse DB (Before)	0.00	2.60	1.11
Horse DB (After)	0.07 (+0.07)	10.15 (+7.55)	0.47 (-0.64)

The increase in PSNR and the decrease in RMSE across all datasets (average over 10 samples) indicate that our method is recovering real structural information, although it is very coarse.

Ablation Study

- ▶ We perform the same inversion method against the same models quantized with round-to-nearest (RTN) rounding.
- ▶ Since RTN does not use calibration data, we compare our method's samples to the mean of the training data instead.

Table: Results from Ablation Study

	SSIM $\uparrow$	PSNR $\uparrow$	NRMSE $\downarrow$
Olivetti Faces (Before)	0.00	2.89	1.12
Olivetti Faces (After)	0.00	3.90 (+1.01)	1.00 (-0.12)
Chest X-Ray (Before)	0.00	2.77	1.11
Chest X-Ray (After)	0.00	3.64 (+0.87)	1.00 (-0.11)
Horse DB (Before)	0.00	2.57	1.10
Horse DB (After)	0.00	3.38 (+0.81)	1.00 (-0.10)

- ▶ We notice 0.00 SSIM on all datasets, implying that the generated samples do not look like the calibration data at all.

Discussion & Limitations

- ▶ Even though the SSIM in the main experiment is quite low, we observe empirical evidence of the leakage channel. Furthermore, the SSIM in the ablation study is 0.00 on all datasets, which empirically shows that AdaRound is likely to be the source of the leakage channel.
- ▶ The main limitation is that our method only reconstructs samples that look like the mean of the training data. It does not reconstruct individual samples as-is. Furthermore, the SSIM we obtained is rather low and there is much room for further improvement.
- ▶ Another limitation is that to perform this attack, the weight perturbation matrix is required. Typically this means needing to obtain both the unquantized and quantized models.

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Method

- ▶ To reconcile AdaRound's assumption that the gradient term in Equation 5 can be ignored, and Central Flows' evidence that the gradient is nonzero even for pretrained converged models, we propose a modified version of AdaRound to incorporate awareness of the local gradient.
- ▶ In order to minimize

$$\mathbb{E} \left[g^{(w)^T} \Delta W \right], \quad (9)$$

we can attempt to force ΔW to have the same sign as $-g^{(w)^T}$.

- ▶ We do this by finetuning the model on a small amount of data then taking the EMA of the model during finetuning as W'_{avg} . We add the Binary Cross Entropy (BCE) between the current rounding directions and $\text{sign}(W'_{\text{avg}} - W)$ as a penalty term to AdaRound's optimization objective.

Algorithm for EMA Finetuning

Algorithm 3 Gradient-Aware Adaptive Rounding (Finetuning)

Require: $\mathbf{W}$, X (calibration data), η (learning rate), B (batch size)

Ensure: EMA of models during finetuning $\mathbf{W}'_{\text{avg}}$

- 1: $\mathbf{W}' \leftarrow \mathbf{W}$
 - 2: $\mathbf{W}'_{\text{avg}} \leftarrow \mathbf{W}$
 - 3: $N \leftarrow \frac{X}{B}$ ▷ Number of Iterations
 - 4: $\alpha \leftarrow \frac{2}{N+1}$ ▷ EMA Smoothing Factor
 - 5: **for** each batch x_b in X **do**
 - 6: $\hat{x}_b \leftarrow \mathbf{W}'(x_b)$ ▷ Forward pass
 - 7: $\mathcal{L} \leftarrow \text{CrossEntropyLoss}(\hat{x}_b, x_b)$
 - 8: $\mathbf{W}' \leftarrow \mathbf{W}' - \eta \cdot \nabla \mathcal{L}$ ▷ Backward pass
 - 9: $\mathbf{W}'_{\text{avg}} \leftarrow \alpha \cdot \mathbf{W}'_{\text{avg}} + (1 - \alpha) \cdot \mathbf{W}'$ ▷ Update EMA
 - 10: **end for**
 - 11: **return** $\mathbf{W}'_{\text{avg}}$
-

Proposed Method Objective

Our proposed method is essentially a modification to AdaRound's objective in Equation 8:

$$\arg \min_{\mathbf{V}} \left(\|\mathbf{W}_X - \hat{\mathbf{W}}_X\|_F^2 + f_{\text{reg}}(\mathbf{V}) + \underbrace{\psi \cdot \text{BCE} \left(h(\mathbf{V}_j), \text{sign}(\mathbf{W}' - \mathbf{W}) \right)}_{\text{Proposed addition}} \right) \quad (10)$$

Experiment Setup

We evaluate the proposed method on our TinyStories models with the following hyperparameters and components.

Table: Implementation Details

Hyperparameter/Component	Value
Finetuning Data	2% of Training Data
Learning Rate	3e-4
Optimizer	AdamW [29]
Learning Rate Scheduler	CosineAnnealingLR [30]
ψ	100.0

Results: Main Experiment

- ▶ We observe **neutral to negative** results using the proposed method.
- ▶ Note that the two models had different testing data (TinyStories v1 and TinyStories v2, respectively).

Table: Perplexity Evaluation of Proposed Method

Type of Quantization	TinyStories V1	TinyStories V2
None (Unquantized Model)	11.9006	8.0994
Round-to-Nearest	11.9219	8.1224
AdaRound	11.9216	8.1177
Proposed Method	11.9216	8.1179

Ablation Study

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Summary

Future Work

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